USN | | | | | | | | | | [22CSE553]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

INDUSTRIAL AI

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	List different types of banking sectors available to fulfill the needs of a wide variety of customers.	L1	CO1	**02**
	b.	Define Alpha and beta returns in the capital asset pricing model.	L1	CO2	**02**
	c.	State the use of "Pydicom" python library.	L1	CO3	**02**
	d.	Define Intelligent maintenance in manufacturing.	L1	CO4	**02**
	e.	Define Supply chain Management.	L1	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Describe the different stages in CRISP –DM model.	L2	CO1	**05**
		OR			
	b.	Discuss the application of AI in banking sector.	L2	CO1	**05**
3.	**a.**	Explain the job of investment bankers in IPO.	L2	CO2	**05**
		OR			
	b.	Outline the challenges in preparing a clustering model for Investment banking.	L2	CO2	**05**
4.	**a.**	Outline the python libraries available in Radiology.	L2	CO3	**05**
		OR			
	b.	Discuss the initiatives taken by IBM Watson for advancements in healthcare using AI.	L2	CO3	**05**
5.	**a.**	Outline the Python libraries available in Pathology.	L2	CO3	**05**
		OR			
	b.	Discuss the objectives of Google Mind/Deep mind for AI in health care.	L2	CO3	**05**
6.	**a.**	Explain how AI Transforms Maintenance Strategies in Manufacturing.	L2	CO4	**05**
		OR			
	b.	Outline the challenges in Implementing AI for Quality Control.	L2	CO4	**05**
7.	**a.**	Outline the importance of Supply Chain Management.	L2	CO5	**05**
		OR			
	b.	Discuss the applications of AI in Educational & Research Sector.	L2	CO6	**05**
		PART-C			**6 X 10=60**
8.	**a.**	With a neat diagram explain the technical stack required to develop an AI application.	L2	CO1	**10**
		OR			

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	b.	Outline Time Series Analysis, and how does the ARIMA model contribute to its effectiveness? Provide a detailed explanation of the ARIMA model, including real-world examples that demonstrate its applications in time series forecasting.	L2	CO1	**10**
9.	**a.**	Summarize and explain the different measures to take care in the process of Auto Syndication for new issues.	L2	CO2	**10**
		OR			
	b.	Explain the following: **i.** Investment Policy statement. **ii.** Markowitz mean-variance model.	L2	CO2	**10**
10.	**a.**	Explain the application of AI in pharmacy.	L2	CO3	**10**
		OR			
	b.	Discuss Key Focus Areas of Google Health and DeepMind in Healthcare.	L2	CO3	**10**
11.	**a.**	Explain how the traditional maintenance can be replaced by AI intelligent maintenance in manufacturing. Briefly describe the advantages and challenges in AI intelligent maintenance in manufacturing.	L2	CO4	**10**
		OR			
	b.	Discuss the Key Components of Machine Learning-Based Demand Planning.	L2	CO4	**10**
12.	**a.**	Summarize the role of AI on the broad cases of: **i.** AI in Procurement. **ii.** AI in Banking Sector and Insurance Industry.	L2	CO5	**10**
		OR			
	b.	Summarize the role of AI on the broad cases of: **i.** Supply Chain Management. **ii.** Educational and Research Field.	L2	CO5	**10**
13.	**a.**	Explain the ethical issues in adopting AI for any Industry.	L2	CO6	**10**
		OR			
	b.	Discuss the various opportunities available for adopting AI in Healthcare sector.	L2	CO6	**10**